# Safety Verification of Chiron IR using Constrained Horn Clauses

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## Introduction

Program verification provides mathematical guarantees that software behaves correctly, unlike testing, which can find bugs but never prove their absence. While the Chiron framework offers support for dynamic analysis techniques such as fuzzing, symbolic execution, and spectrum-based fault localization, it currently lacks the capability to *prove* that a program satisfies a given specification for *all* possible inputs.

This project bridges the gap between testing and verification in Chiron by implementing a Property Directed Reachability (PDR)-based safety verification engine, that given a program and a set of properties on the program state, either guarantees that they hold or provides a counterexample for the same. We target properties natural to turtle graphics programs, such as spatial bounds on the turtle's position and invariants over program variables.

## Base Flow & Tools Used

### Overview of the pipeline

The pipeline takes a source turtle program and a set of safety properties as input. Using Chiron's API, the program is translated to Chiron-IR and its relevant user and state variables are extracted. The program semantics are then encoded as Constrained Horn Clauses (CHCs), and combined with the properties expressed in CHC form. This CHC system is given to Z3's `SPACER` engine, which checks whether an **unsafe** state is reachable. A `sat` result yields a concrete counterexample, while `unsat` proves that the property holds.

### Features

| Category | Option | Description |
|---|---|---|
| Mode | universal | Checks the property over all possible values of the user input variables, with the turtle initially at the origin and `pendown=True`. |
| | default | Checks the property with all user variables initialized to 0, with the turtle initially at the origin and `pendown=False`. |
| | specific | Similar to **default**, except that the user may explicitly assign values to some or all user input variables. |
| Property Scope | all | Interprets the safety guarantee as required to hold at every point during program execution. |
| | terminating | Interprets the safety guarantee as required to hold only at program termination. |

Table 1. Verification modes and property scopes supported by the verifier.

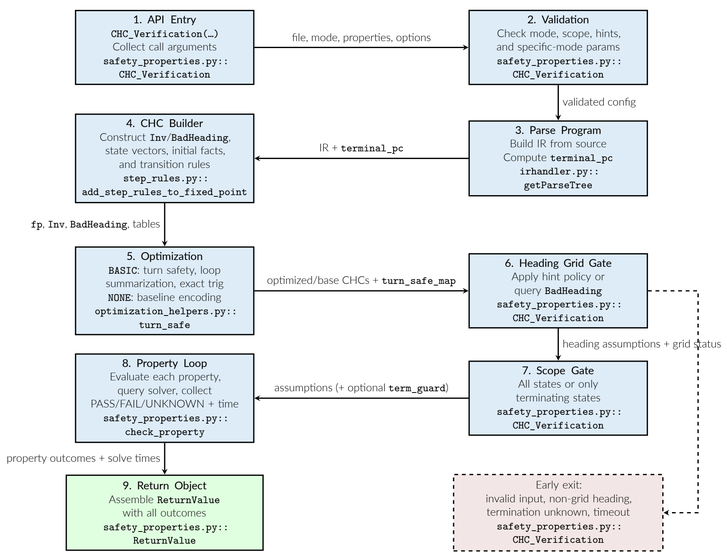



Figure 1. Main verification pipeline showing stage responsibilities and data artifacts flowing through parsing, CHC construction, heading/scope controls, and final result aggregation.

## Design Choices & Optimizations

### Design Choices

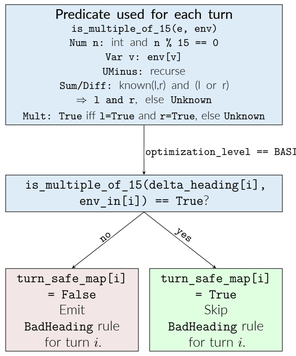
This section summarizes the principal modeling and solver choices adopted in the verifier.

- **CHC semantics with Z3 Fixedpoint/SPACER:** Program behavior is encoded as CHCs and solved using Z3 `Fixedpoint` with `engine='spacer'` (an IC3/PDR-style Horn-clause engine), with safety posed as `Exists(s, Inv(s) & Not(P(s)))`.
- **Single global invariant schema:** A unified relation `Inv(pc, xcor, ycor, heading, pendown, user_vars, loop_counters)` is used across initialization, transition rules, and property queries.
- **Heading-lattice geometry model:** Heading is normalized to `[0,360)` and constrained to `{0,15,…,345}`, with violations represented explicitly via `BadHeading`.

| Design Choice | Benefit | Tradeoff |
|---|---|---|
| CHCs solved by Z3 SPACER | Supports unbounded safety checking through inductive invariant synthesis and returns counterexample witnesses on violation. | Performance remains sensitive to nonlinear arithmetic obligations; solver outcomes may be `UNKNOWN`. |
| Full state (pc+counters+vars) | Encodes control-state and data-state dependencies in one relation: `pc` for control/termination, `xcor/ycor/heading` for geometric predicates, and loop counters for repeat constructs. | Properties over a strict subset of variables still incur full-arity reasoning, increasing solver cost. |
| Strict heading lattice + `BadHeading` | Avoids direct transcendental `sin/cos` constraints in SPACER by using a finite 15-degree trigonometric lookup table (precomputed in Python) with explicit `BadHeading` obligations. | Lookup expansion must remain bounded to control CHC size; failure to prove grid preservation yields conservative `UNKNOWN` outcomes. |

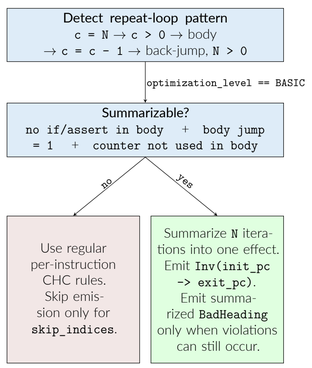
### Optimizations

#### Turn-Safe Heading Inference




```
if :u > 0 [
  :t = 30
] else [
  :t = 45
]
left :t
forward 20
```

#### Repeat-Loop Summarization




## Outputs & Perf Analysis

### Verifier Outputs

Each `CHC_Verification()` call returns a verdict per property:

```
$ CHC_Verification("square.tl", "default", […])
[PASSED] Property 'pen_up' is an invariant.
[FAILED] Property 'x_pos': counterexample found.
[UNKNOWN] Property 'heading': solver timed out.
build: 0.094 s  solve: 0.03 s, 1.74 s, 60.0 s
```

### Correctness Tests

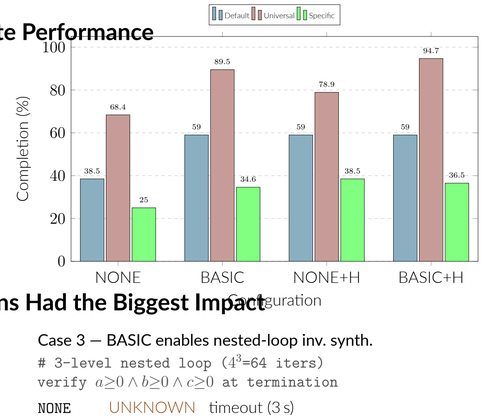
The correctness test suite has, **206 tests** over **64 fixture programs** for 3 verification modes,

| Mode | Tests | Coverage |
|---|---|---|
| Default | 48 | Arithmetic, geometric, pen, heading, nested loops |
| Specific | 70 | Parameterised init; boundary-sensitive outcomes |
| Universal | 90 | Symbolic initial state; relational & terminating props |

6 tests skipped (timeout-prone trig edge cases). **All 200 active tests pass.**

### Aggregate Performance

| Mode | NONE | BASIC | NONE+H | BASIC+H |
|---|---|---|---|---|
| Default | 38.5% | 59.0% | 59.0% | 59.0% |
| Universal | 68.4% | 89.5% | 78.9% | **94.7%** |
| Specific | 25.0% | 34.6% | 38.5% | 36.5% |
| **Overall** | 41.9% | 58.1% | 56.6% | **60.5%** |




### Where Optimisations Had the Biggest Impact

#### Case 1 — BASIC unlocks trig programs

```
# repeat 10 [ forward 10; right 90 ]
verify heading ∈ {0,90,180,270}
```

NONE UNKNOWN timeout (60 s)
BASIC PASSED solve **0.39 s**

*Why:* NONE encodes **`xcor/ycor`** via **`sin/cos`** of a symbolic heading — Z3 cannot decide the nonlinear arithmetic. BASIC detects all turns are multiples of 15°, switches to an *integer heading grid*, eliminating sin/cos entirely.

#### Case 2 — Hint trivialises heading-grid

```
# repeat 20 [ right 15; left 15 ]
verify heading ∈ 15°·ℤ
```

NONE UNKNOWN timeout (60 s)
BASIC UNKNOWN timeout (60 s)
NONE+H PASSED solve **0.107 s**

*Why:* Both NONE and BASIC must *derive* heading-grid membership via CHC traversal — the 24-way disjunction always times out. The hint **`heading_on_grid_always`** injects it as an axiom, collapsing the proof to trivial modular arithmetic.

#### Case 3 — BASIC enables nested-loop inv. synth.

```
# 3-level nested loop (4^3=64 iters)
verify a≥0 ∧ b≥0 ∧ c≥0 at termination
```

NONE UNKNOWN timeout (3 s)
BASIC PASSED solve **2.29 s**
BASIC+H PASSED solve **0.163 s**

*Why:* NONE generates **`BadHeading`** CHC rules for every state, expanding the search space beyond Spacer's reach. BASIC's static turn-safety pass proves no **`BadHeading`** rules are needed, letting Spacer converge. Adding the hint gives a further 14× speedup.

#### Case 4 — BASIC cuts solve time 220× (Universal)

```
# repeat 100 [ :x = :x + 1 ]
verify x ≤ 100 at termination
```

NONE PASSED solve **3.54 s**
BASIC PASSED solve **0.016 s**

*Why:* NONE runs a heading-grid safety query first — even with no turns, Spacer traverses all reachable states, caching lemmas that hurt the main proof. BASIC proves vacuous turn-safety and *skips that query entirely*, keeping the solver state clean.

## Future Work

- **Property-aware state projection:** Project the state space onto only the variables relevant to a given property, shrinking the CHC encoding and reducing solver load.
- **Stronger turn-safety check:** The current heading-grid check conservatively requires both operands of a product to be multiples of 15°. Incorporating richer program invariants will reduce unnecessary `UNKNOWN` verdicts.
- **Recursive loop summarization:** For perfectly nested affine loops, summarize the inner loop first into a single affine state transformer, then summarize the outer loop with it — drastically reducing CHC clause counts for deep nesting.
- **User-specified focus variables (`focus_vars`):** Let users name the variables most relevant to a property so the solver reasons over a smaller slice of the state.
- **Semantic property helpers:** Replace raw property strings with domain-specific helpers such as `heading_in(range)`, `position_in_box(...)`, `pen_is_down()` for a friendlier API.
- **Variable domain declarations:** Extend `specific` mode to accept intervals or sets of initial values rather than a single concrete assignment.

## References

[1] Claude E. Shannon. A mathematical theory of communication. *The Bell System Technical Journal*, 27(3):379–423, 1948. doi:10.1002/j.1538-7305.1948.tb01338.x.